

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)

SCALA FOR DS

Practical - 14,15,16,17 & 18

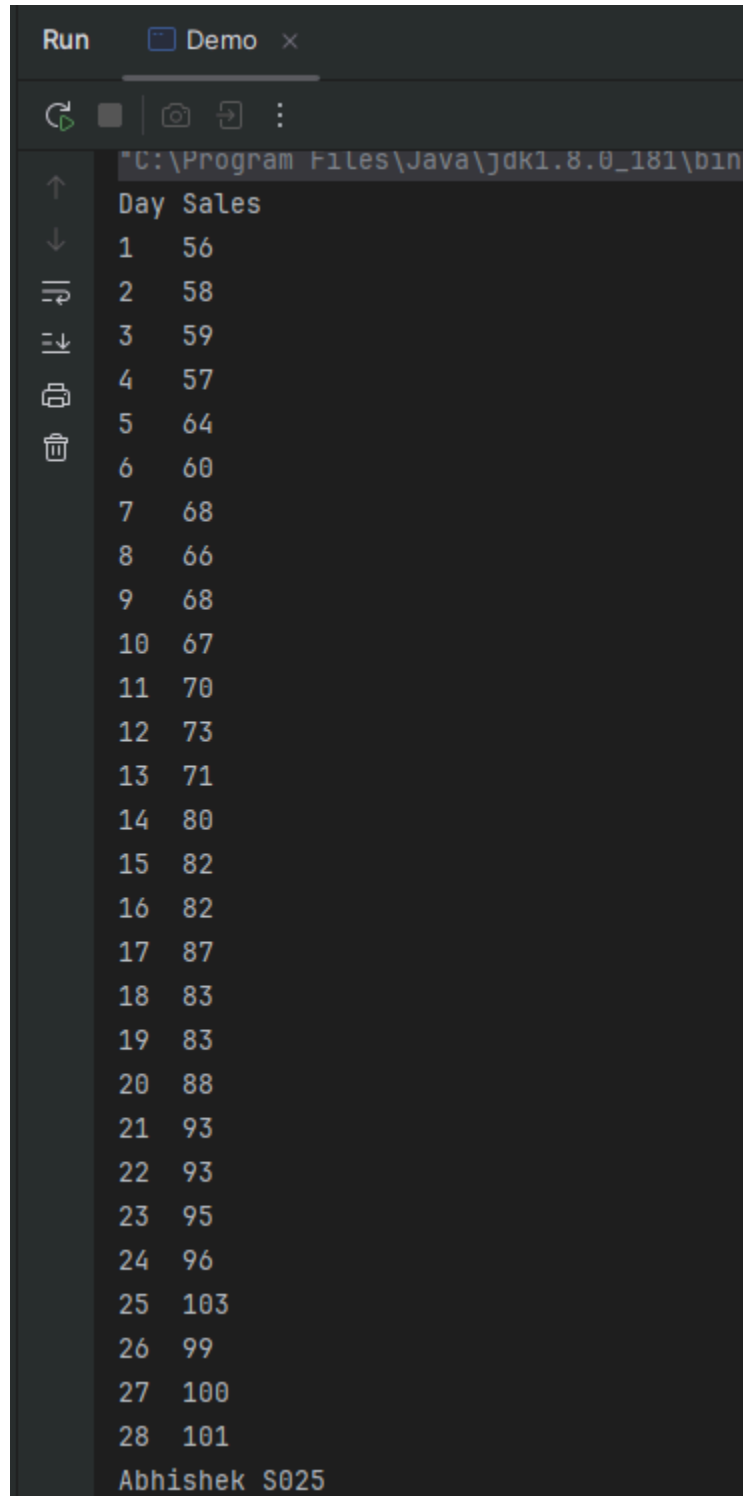
Roll No. : S025	Name: Abhishek Pal
Class : S.Y.I.T	Batch : 1
Date of Assignment : 11/08/2026	Date/Time of Submission : 11/08/2026

14) Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

CODE :

```
build.sbt  Demo.scala x
1  import scala.util.Random
2  object Demo extends App {
3      println("Day\tSales")
4      for (day <- 1 to 28) {
5          val sales = 50 + day * 2 + Random.nextInt(10) - 5
6          println(s"$day\t$sales")
7      }
8      println("Abhishek S025")
9  }
```

OUTPUT :



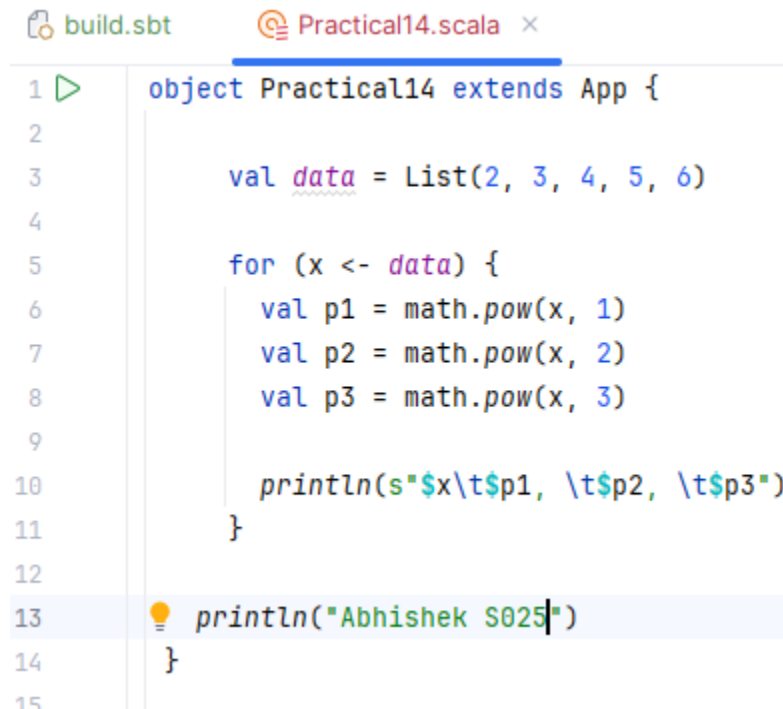
The screenshot shows a Java IDE console window titled "Run" with a tab labeled "Demo". The console output displays a table with two columns: "Day" and "Sales". The data is as follows:

Day	Sales
1	56
2	58
3	59
4	57
5	64
6	60
7	68
8	66
9	68
10	67
11	70
12	73
13	71
14	80
15	82
16	82
17	87
18	83
19	83
20	88
21	93
22	93
23	95
24	96
25	103
26	99
27	100
28	101

At the bottom of the console output, the text "Abhishek S025" is displayed.

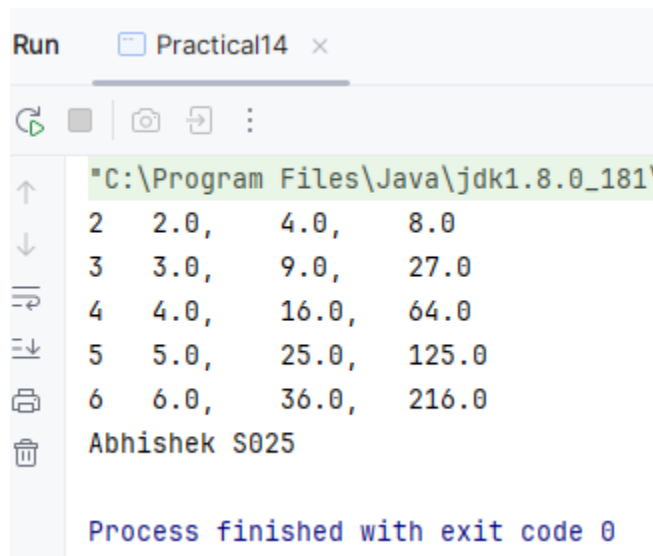
15) Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.

CODE :



```
build.sbt  Practical14.scala x
1  object Practical14 extends App {
2
3      val data = List(2, 3, 4, 5, 6)
4
5      for (x <- data) {
6          val p1 = math.pow(x, 1)
7          val p2 = math.pow(x, 2)
8          val p3 = math.pow(x, 3)
9
10         println(s"$x\t$p1, \t$p2, \t$p3")
11     }
12
13     println("Abhishek S025")
14 }
15
```

OUTPUT :



```
Run  Practical14 x
C:\Program Files\Java\jdk1.8.0_181\
2  2.0,    4.0,    8.0
3  3.0,    9.0,   27.0
4  4.0,   16.0,   64.0
5  5.0,   25.0,  125.0
6  6.0,   36.0,  216.0
Abhishek S025

Process finished with exit code 0
```

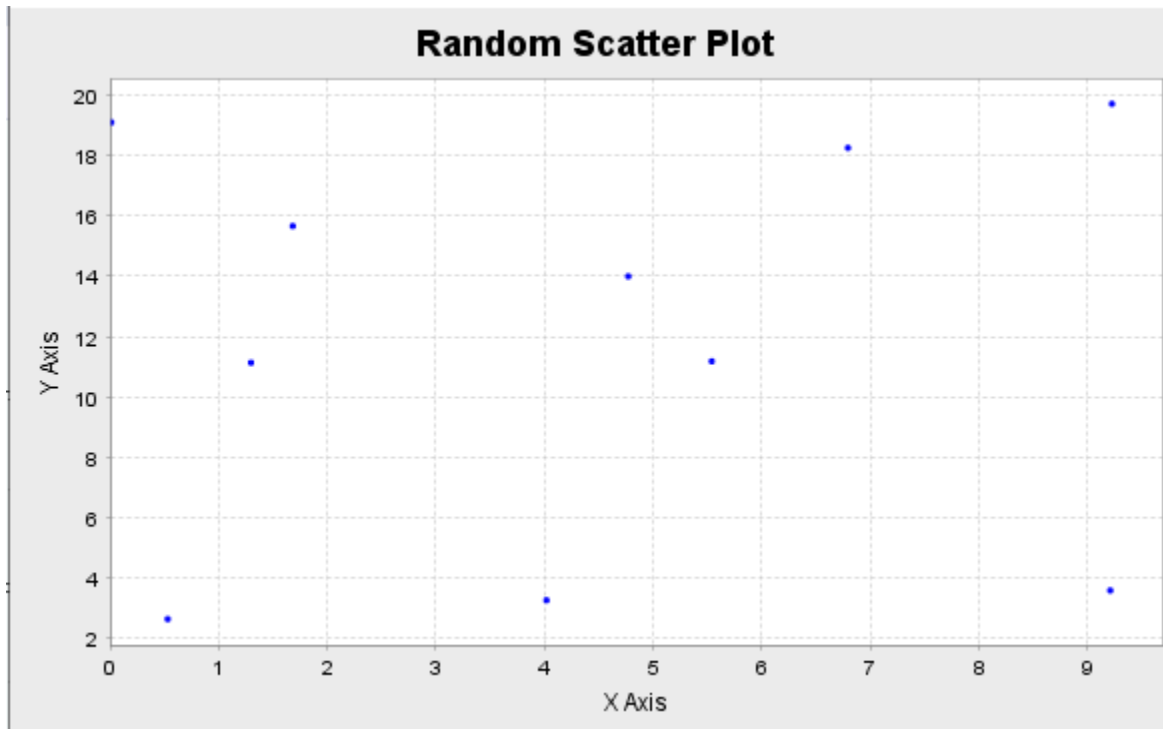
16) Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

CODE :

```
build.sbt x Practical14.scala
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.13.12"
4 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
5
6 libraryDependencies += "org.scalanlp" %% "breeze-viz" % "2.1.0"
7 lazy val root = (project in file(".")) .settings( name := "Practical14"
8 )
9
```

```
1 import breeze.plot._
2 import scala.util.Random
3
4 object Pract16 extends App {
5   val x = List.fill(10)(Random.nextDouble() * 10)
6   val y = List.fill(10)(Random.nextDouble() * 20)
7   println(x)
8   println(y)
9   val f = Figure()
10  val p = f.subplot(0)
11
12  p += plot(x, y, '.', colorcode = "blue")
13
14  p.xlabel = "X Axis"
15  p.ylabel = "Y Axis"
16  p.title = "Random Scatter Plot"
17
18  f.refresh()
19
20 }
```

OUTPUT :



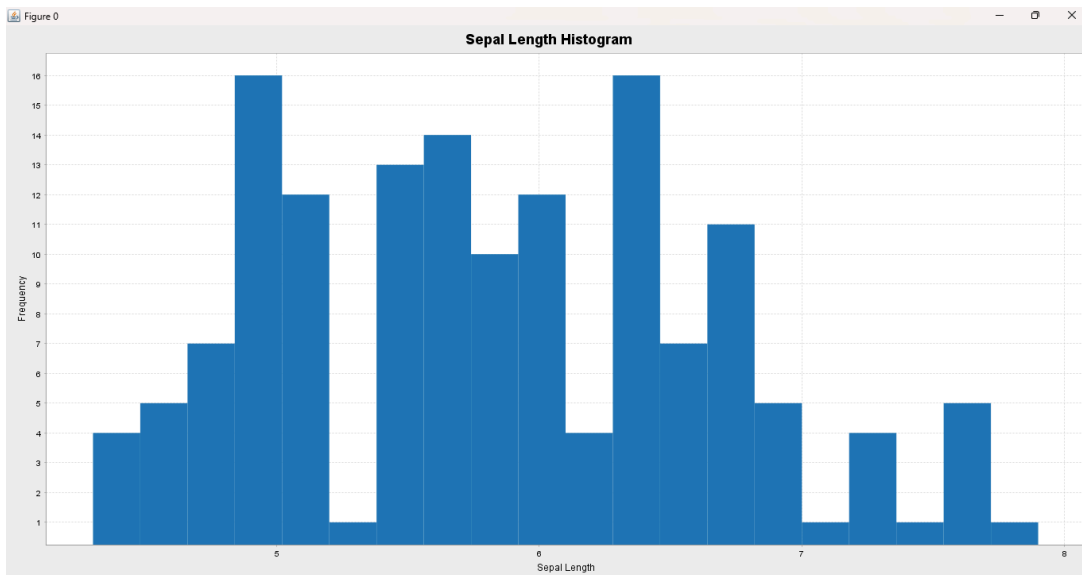
```
"C:\Program Files\Java\jdk1.8.0_341\bin\java.exe" ...  
List(5.0347772553468, 0.06733192086907258, 0.5803521987787008, 1.291383683038403, 0.36454589972769624, 0.3473192642225842, 7.643900153538873, 8.009843954,  
List(8.864737524676338, 3.8931580056058213, 12.17358345279992, 15.865509070096449, 8.93215585265915, 10.088288987554787, 14.979717256563731, 1.51844985118)
```

17) Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.

CODE :

```
1
2  import breeze.plot._
3  import scala.io.Source
4
5  object Pract17 extends App {
6    val file = Source.fromFile("src/main/resources/iris.csv")
7    val data = file.getLines()
8      .drop(1)
9      .map(line => line.split(",")(0).toDouble)
10     .toList
11    file.close()
12    val fig = Figure()
13    val plt = fig.subplot(0)
14    plt += hist(data, bins = 20)
15    plt.title = "Sepal Length Histogram"
16    plt.xlabel = "Sepal Length"
17    plt.ylabel = "Frequency"
18    fig.refresh()
19  }
20
```

OUTPUT :



18) Plot a line graph for a dataset showing a trend over time.

CODE :

```
import breeze.plot._
import scala.io.Source

> object Demo extends App {

    val file = Source.fromResource("iris.csv")

    val x = file.getLines()
        .drop(1)
        .map(line => line.split(",")(0).toDouble)
        .toList

    file.close()

    val y = x.indices.map(_._2.toDouble).toList

    val fig = Figure()
    val p = fig.subplot(0)

    p += plot(y, x)

    p.xlabel = "Record Number"
    p.ylabel = "Sepal Length"
    p.title = "Sepal Length Trend"

    fig.refresh()
}
```

OUTPUT :

